Math 10C — Unit 4: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Monomial Common Factors (Questions 1–13)

1. Write the prime factorization of $20a^3$ and $32a^2$, and determine the greatest common factor of $20a^3$ and $32a^2$.

2. Write the prime factorization of $12xy^4$ and $18x^2y^3$, and determine the greatest common factor of $12xy^4$ and $18x^2y^3$.

3. State the greatest common factor of each set of monomials.

(a) $9m,\ 15m$ (b) $8x^2,\ 10x$ (c) $ab^2,\ ab^5$ (d) $mn,\ m^2n^2$

(e) $6x^3,\ 9x^2$ (f) $5xyz,\ 15rst,\ 25def$ (g) $-6pq^2,\ 12p^2q$ (h) $-12x^3z^5,\ -18x^2z^3$

(i) $6a^2b,\ 10ab,\ 4a^2$ (j) $12mn,\ 18np,\ 24mnp$ (k) $-3a^2b,\ -6a^2b^3,\ -9ab^2$ (l) $-14pqr^2,\ -21p^2q,\ -35q^2r$

4. Complete the factoring in each case.

(a) $15a + 20b =$ _____$(3a + 4b)$ (b) $9p^2 - 6p =$ _____$(3p - 2)$

(c) $4xy + 6xz =$ _____$(2y + 3z)$ (d) $8x^2 + 16x + 24 =$ _____$(x^2 + 2x + 3)$

(e) $15cde - 9cd =$ _____$(5e - 3)$ (f) $6y^3 - 18y =$ _____$(y^2 - 3)$

5. In each case, the greatest common factor has been removed. Complete the factoring.

(a) $4a^2 + 16a = 4a(a +$ _____$)$ (b) $24p - 15q = 3($_____$ - 5q)$

(c) $8x^3 - 12x^2 = 4x^2($_____$)$ (d) $6a^2b + 9a^3b^2 = 3a^2b($_____$)$

(e) $-20x^2y - 15x^2y^2 = -5x^2y($_____$)$ (f) $24xm^2n^3 - 18mn^2 - 6mn = 6mn($_____$)$

6. Factor the following polynomials by removing the greatest common factor.

(a) $8m + 8n$ (b) $9xy^2 + 81$ (c) $21pq - 7$ (d) $10c + 15d$

(e) $xz + z$ (f) $8x^2 - 12x$ (g) $12ab - 16ac$ (h) $54y^2 - 81y^5$

7. Factor the following polynomials.

(a) $15x - 10y + 25z$ (b) $8pq + 12pr - 20p$ (c) $t^4 + t^3 + t^2$

(d) $7x^2 - 14xy - 28xz$ (e) $6abc - 3abd + 9abe$ (f) $21a^2b^2 + 28a^3b^2 - 42a^2b^3$

8. In each of the following: (i) simplify the expression by combining like terms; (ii) factor the resulting polynomial.

(a) $6x^2 - 3x + 5 - 3x^2 + 7x - 5$ (b) $8 - 3y + 4y^2 - 6y + 2y^2 - 8$

(c) $ab^3 - 3a^3b + 6a^2b^2 - 2ab^3 + 5a^3b$ (d) $3(x^2 - 2x) - 5x(x - 3) + 4x^2(x - 1) - 2x$

9. Factor the following polynomials. Expand your answer to verify the factoring.

(a) $30x^3 - 45x^2$ (b) $-12p^3 - 24p^2 - 12p$

10. The surface area of a cylinder is given by the formula $A = 2\pi r^2 + 2\pi rh$, where r is the radius of the base and h is the height of the cylinder.

(a) Calculate the surface area, to the nearest 0.01 cm^2, of a cylinder which has height 16.2 cm and base diameter 14 cm.

(b) Write the formula for A in factored form.

(c) Calculate the surface area of the cylinder, to the nearest 0.01 cm^2, using the factored form of A.

(d) Which method, (a) or (c), is simpler to use?

11. A rocket is launched straight upward from the ground with an initial speed of 40 m/s. The height (h metres) of the rocket above the ground after t seconds can be approximated by the formula $h = 40t - 5t^2$.
(a) Write h in factored form.

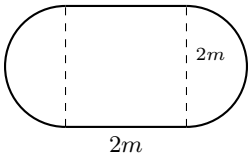
(b) Use the factored form of h to calculate the height for each of the times in the table. Record your answers in the table.

Time (t seconds)	0	1	2	3	4	5	6	7	8
Height (h metres)									

(c) Explain why the height of the rocket after 3 seconds is the same as the height of the rocket after 5 seconds.

(d) Calculate h when $t = 9$. Explain why this has no meaning in the context of the question.

12. A square of side $2m$ cm has semicircles drawn externally on each of two opposite sides, as shown. Find expressions in factored form for



(a) the external perimeter of the shape

(b) the area of the shape

13. Identify and explain the error in each factorization. Then write the correct factored form.
(a) $12x^3 - 8x^2 = 4x(3x^2 - 2x)$ (b) $-6m^2 - 9m = -3m(2m - 3)$ (c) $4xy + 2x = 2x(2y)$

Part B — Multiple Choice and Numerical Response (Questions 14–18)

14. *(Multiple Choice)* $4\pi r^3 + 12\pi r$ is equivalent to
A. $4\pi^2 r^4$ B. $4\pi(r^2 + 3r)$
C. $4\pi r(r^2 + 3)$ D. $4\pi r(r^2 + 3r)$

15. *(Multiple Choice)* One factor of $12x^4 - 8x^3 + 4x^2$ is
A. $12x^4$ B. $4x^2 - 2x$
C. $4x^2 - 8x + 4$ D. $3x^2 - 2x + 1$

16. *(Multiple Choice)* How many of the following three statements are true?
i) $x^2 + xy + x = x(x + y)$
ii) $\frac{1}{4}\pi r^3 + \pi r^2 h = \frac{1}{4}\pi r^2(r + 4h)$
iii) $x^3 + x^5 + x^7 = x^3(1 + x^2 + x^3)$
A. 0 **B.** 1 **C.** 2 **D.** 3

17. *(Numerical Response)* When $x^5y^4 - x^3y^4 + x^7y$ is factored, the greatest common factor has degree A and the remaining polynomial factor has degree B. The value of $A + 2B$ is _____.

Record your answer left to right. | | | | |

18. *(Numerical Response)* When the greatest common factor is removed from the binomial $60x^2 - 40x$, the binomial can be written in the form $ax(bx + c)$. The value of $a + b + c$ is _____.

Record your answer left to right. | | | | |

Part C — Extension: Fractional Common Factors (Questions 19–23)

19. *(Extension)* In each case, a common factor has been removed so that the remaining factor is an integral binomial. Complete the factoring.

(a) $\frac{1}{3}x - x^2 = \frac{1}{3}x(\ -\ )$ (b) $5a^2 + \frac{3}{4}a = \frac{1}{4}a(\ +\ )$ (c) $-\frac{1}{5}y^2 - y = -\frac{1}{5}y(\ +\ )$

20. *(Extension)* Complete the factoring and check mentally by expanding the factored form.

(a) $\frac{1}{7}n - \frac{3}{7} =$ _____$(n - 3)$ (b) $\frac{1}{3}x^3 + \frac{1}{6}x =$ _____$(2x^2 + 1)$ (c) $0.4a^3 - a =$ _____$(2a^2 - 5)$

21. *(Extension)* In each case, remove a common factor so that the remaining factor is an integral polynomial.

(a) $\frac{1}{3} + \frac{1}{3}x$ (b) $\frac{1}{4}m - \frac{5}{4}$ (c) $\frac{2}{3}ah - \frac{3}{4}bh$ (d) $\frac{3}{4}x + 9$

(e) $\frac{5}{2}x^2 - 1$ (f) $z^2 - \frac{1}{3}z + \frac{2}{3}$ (g) $\frac{5}{3}\pi r^3 - 5\pi r^2$ (h) $\frac{1}{4}Ah + \pi r^2 h$

22. *(Extension)* An object is thrown down from a high building. The distance, s metres, travelled by the object in t seconds is given by the formula $s = vt + \frac{1}{2}at^2$.

(a) Express s in factored form.

(b) Use the factored form to calculate the distance travelled in 0.5 seconds if $v = 6$ and $a = 10$.

23. *(Extension)* Identify and explain the error in each of the following polynomial factorizations.

(a) $\frac{1}{4}x^2 - \frac{1}{8}x\ =\ \frac{1}{4}x(x - 2)$ (b) $\frac{1}{4}\pi r^2 - 2\pi r\ =\ \frac{1}{4}\pi r(r - 8\pi)$

Lesson 2 Assignment *Show your work.*

Part A — Binomial Common Factors and Grouping (Questions 1–5)

1. Factor the following polynomials by removing the greatest common factor.

(a) $(4x)(x+6)-(5)(x+6)$ (b) $9y(x+3)+4(x+3)$ (c) $x(x+2)-(x+2)$

(d) $(m+n)(x)-(m+n)(y)$ (e) $6x(a+b)-12y(a+b)$ (f) $3a(a+7)-6(a+7)$

(g) $15a(b+c)+5d(b+c)$ (h) $x(4x-5)-3(5-4x)$ (i) $5(2-9p)+4p(-2+9p)$

2. Factor the following polynomials by grouping.

(a) $x^2+3x+7x+21$ (b) $x^2+4x+12x+48$ (c) $m^2-6m+3m-18$

(d) $a^2-8a-6a+48$ (e) $x^2-12x-5x+60$ (f) $t^2+9t-4t-36$

3. Factor the following polynomials by grouping.

(a) $3x^2+3x+4x+4$ (b) $4x^2+x+8x+2$ (c) $2m^2+8m+3m+12$

(d) $10b^2-15b-8b+12$ (e) $3a^2-9a-2a+6$ (f) $7x^2+3x-35x-15$

(g) $25+5p+5p+p^2$ (h) $21-3y-7y+y^2$ (i) $b^2+bx+by+xy$

4. Factor the following polynomials by grouping. *(Rearrange the terms first.)*

(a) $pq+x^2-px-qx$ (b) $6b^2+3a-18b-ab$ (c) $5x^2+21y^2-15xy^2-7x$

5. Identify and explain the error in each factorization. Then write the correct factored form.

(a) $m^2-5m-40+8m \;=\; m(m-5)-8(5-m) \;=\; (m-5)(m-8)$

(b) $2x^2 + 6x - 5x - 15 = 2x(x + 3) - 5(x - 3)$, so it cannot be factored by grouping.

(c) $3a(a - 4) + (a - 4) = (a - 4)(3a)$

Part B — Multiple Choice and Numerical Response (Questions 6–9)

6. *(Multiple Choice)* One factor of $pq - 3ps - 15rs + 5rq$ is

A. $(p - 5r)$ **B.** $(3s - p)$
C. $(q - 3s)$ **D.** $(q + 3s)$

7. *(Multiple Choice)* $p(p - 5) + q(5 - p)$ is equivalent to

A. $(p - 5)(p + q)$ **B.** $(p - 5)(p - q)$
C. $(5 - p)(p - q)$ **D.** $(5 - p)(p + q)$

8. *(Multiple Choice)* Consider $4a^2 + 3a + 20a + c$, where c is a constant. If $a + 5$ is a factor, then the other factor must be

A. $4a + 3$ **B.** $a + 3$
C. $4a + 15$ **D.** $a + 15$

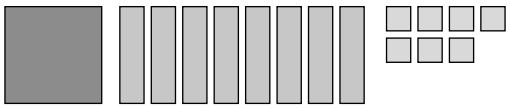
9. *(Numerical Response)* The polynomial $3x^2 + 2x + 15x + p$ has a factor of $(3x + 2)$. The value of p is _____.

Record your answer left to right. | | | | |

Lesson 3 Assignment *Show your work.*

Part A — Algebra Tiles and Sum–Product Pairs (Questions 1–4)

1. Consider the collection of algebra tiles shown below.



(a) Write the polynomial expression represented by the tiles.

(b) Arrange the tiles into a rectangle, and state the length and width.

(c) Write the polynomial in factored form.

2. Complete each table by finding two integers with the given sum and given product.

Sum	Product	Integers
(a) 7	12	
(b) 9	8	
(c) 13	40	
(d) −13	40	

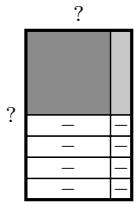
Sum	Product	Integers
(e) 10	9	
(f) −9	14	
(g) −19	48	
(h) −20	96	

3. Complete the table to find two integers with the given sum and the given product.

	Sum	Product	Integers
a)	5	−24	
b)	−5	−24	
c)	−2	−24	

	Sum	Product	Integers
d)	4	−45	
e)	−13	36	
f)	0	−25	

4. The rectangle below is made of algebra tiles (key in Section 1 of the notes).



(a) Write the trinomial the tiles represent.

(b) State the length and width, and write the trinomial in factored form.

(c) Explain why the unit tiles are negative.

Part B — Factoring $x^2 + bx + c$ (Questions 5–14)

5. Complete the following.

(a) $x^2 + 9x + 20 = (x + 4)(x + ___)$ (b) $x^2 + 11x + 10 = (x + ___)(x + ___)$ (c) $x^2 - 11x + 18 = (x - 2)(x - ___)$

(d) $t^2 - 16t + 15 = (t - ___)(t - ___)$ (e) $z^2 + 12z + 20 = (z + 2)(_____)$ (f) $b^2 - 14b + 24 = (b - 2)(_____)$

6. Factor the following. Verify (a) by expanding.

(a) $x^2 + 5x + 4$ (b) $x^2 - 5x + 4$

(c) $x^2 - 12x + 27$ (d) $x^2 - 13x + 30$

7. Factor where possible.

(a) $x^2 + 13x + 12$ (b) $x^2 + 12x + 13$ (c) $y^2 - 13y + 30$

(d) $y^2 + 19y + 48$ (e) $f^2 - 11f + 24$ (f) $p^2 - 17p + 30$

(g) $c^2 + 10c + 15$

8. Factor the following trinomials.

(a) $x^2 - 3x - 10$ (b) $x^2 - 3x - 40$

(c) $x^2 + 3x - 40$ (d) $x^2 - 7x - 18$

9. Factor where possible.

(a) $x^2 - 13x + 22$ (b) $x^2 - 4x - 21$ (c) $x^2 + 6x - 27$ (d) $x^2 - 5x + 14$

(e) $x^2 - 5x - 14$ (f) $x^2 - 12x + 36$ (g) $m^2 + 23m + 42$ (h) $a^2 - 19a + 48$

10. Factor completely.

(a) $2x^2 + 10x + 8$ (b) $3x^2 - 42x + 144$ (c) $-2a^2 - 26a - 84$

(d) $4x^2 - 16x + 12$ (e) $ax^2 - 11ax + 30a$ (f) $-6a^4 + 66a^3 - 180a^2$

11. Factor completely.
(a) $2x^2 + 16x + 30$ (b) $4x^2 - 24x - 28$ (c) $5x^2 - 30x + 25$

(d) $-2a^2 + 4a + 96$ (e) $b^2x^2 - 6b^2x - 40b^2$ (f) $3x^3 + 3x^2 - 60x$

12. A rectangular soccer field has an area of $x^2 + 17x + 70$ square metres.
(a) Factor $x^2 + 17x + 70$ to find binomials representing the length and width of the field.
(b) If $x = 5$, determine the length and width of the field.

13. In the following, each letter represents a different whole number from 1 to 9.

$$x^2 + 3x - 4 = (x + O)(x - B) \qquad x^2 + 2x - 15 = (x - L)(x + A) \qquad x^3 + 4x^2 - 12x = x(x - I)(x + G)$$

$$9x^2 - 126x + 441 = T(x - N)^2 \qquad -9x^2 + 9x + 648 = -9(x + S)(x - T)$$

Determine the value of each letter and hence identify the fictional character represented by the following code.

(1) (2) (3) (1) (4) (1) (5) (6) (6) (2) (7) (8)

14. Identify and explain the error in each factorization. Then write the correct factored form.
(a) $x^2 - 5x - 14 = (x - 7)(x - 2)$ (b) $x^2 + 7x + 12 = (x + 12)(x + 1)$ (c) $2x^2 + 10x + 12 = (x + 2)(x + 3)$

Part C — Two Variables (Questions 15–16)

15. Factor each trinomial completely.
(a) $x^2 + 16xy + 63y^2$ (b) $x^2 + 8xy - 20y^2$ (c) $a^2 - 14ab + 49b^2$

(d) $p^2 - 13pq + 12q^2$ (e) $x^2 + xy - 90y^2$ (f) $x^2 - 22xy - 75y^2$

16. Factor completely.
(a) $3x^2 - 51xy + 216y^2$ (b) $2b^2 - 12bv - 320v^2$ (c) $4c^2 + 8cd - 320d^2$

Part D — Multiple Choice and Numerical Response (Questions 17–23)

17. *(Multiple Choice)* Which of the following is **not** a factor of $2m^2 - 22m + 60$?
A. $m - 5$ **B.** $m - 6$ **C.** $m - 3$ **D.** 2

18. *(Multiple Choice)* For which of the following trinomials is $(a + 6)$ **not** a factor?
A. $a^2 + 7a + 6$ **B.** $a^2 + 11a + 30$ **C.** $a^2 + 12a + 36$ **D.** $a^2 + 13a + 30$

19. *(Multiple Choice)* When factored, the trinomials $x^2 - 9x + 20$ and $x^2 + 3x - 40$ have one binomial factor in common. This factor is
A. $x - 5$ **B.** $x + 5$ **C.** $x - 4$ **D.** $x + 8$

20. *(Multiple Choice)* One factor of $-m^3 - 3m^2 + 10m$ is
A. $m - 2$ **B.** $m + 2$ **C.** $m - 5$ **D.** $m + 10$

21. *(Multiple Choice)* One factor of $5x^2 - 5xy - 100y^2$ is
A. $5x$ **B.** $x - 4y$ **C.** $x + 5y$ **D.** $x + 4y$

22. *(Multiple Choice)* The expression $x^2 - 6x + c$ **cannot** be factored (over the integers) if c has the value
A. 10 **B.** -7 **C.** 8 **D.** 9

23. *(Numerical Response)* The largest value of b for which $x^2 + bx + 36$ can be factored over the integers is _____.
Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

1. Complete the following by determining the missing factor.
(a) $y^2 - 64 = (y - 8)(_____)$ (b) $d^2 - 169 = (d + 13)(_____)$ (c) $r^2 - s^2 = (r - s)(_____)$

2. Factor the following polynomials using a difference of squares. Verify (a) by expanding.
(a) $x^2 - 81$ (b) $x^2 - 1$ (c) $x^2 - 13^2$ (d) $x^2 - 900$

3. Explain how factoring a difference of squares in one variable can be regarded as a special case of factoring trinomials by inspection.

4. Factor where possible.
(a) $p^2 - q^2$ (b) $e^2 - 81$ (c) $1 - j^2$ (d) $f^2 - 49g^2$

(e) $16x^2 - 81$ (f) $25a^2 - 4b^2$ (g) $9x^2 + y^2$ (h) $64c^2 - 121d^2$

(i) $x^4 - 1$ *(Lesson 6 preview)*

5. The surface of a rectangular deck has area $49p^2 - 9q^2$ square feet.
(a) Write expressions for the length and width of the deck.

(b) If the perimeter of the deck is 252 feet, form an equation in p and q and solve for p.

(c) Determine the length and width of the deck if the length is twice the width.

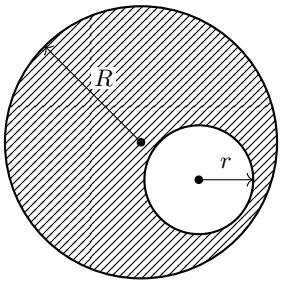
6. Factor.
(a) $5x^2 - 45$ (b) $6a^2 - 24y^2$ (c) $2t^2 + 18s^2$

(d) $3x^2 - 3y^2$ (e) $4a^2b^2 - 64$ (f) $12 - 3p^2q^2$

7. Factor.
(a) $x^2y^2 - 16$ (b) $a^2 - b^2c^2$ (c) $81x^2 - y^2z^2$

(d) $m^2n^2 - p^2q^2$ (e) $16x^2y^2 - 1$ (f) $a^2b^2 - 9c^2d^2$

8. The diagram shows a circle of radius R with a circle of radius r removed.



(a) Write an expression for the shaded area.

(b) Write the expression in (a) in factored form.

(c) Determine the shaded area (as a multiple of π) if $R = 7.5$ and $r = 2.5$. Do not use a calculator.

9. The expression $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$ occurs in physics.
(a) Write the expression in factored form.

(b) Determine the value of the expression when $m = 8$, $v = 50$, and $u = 30$. Do not use a calculator.

10. Farid was showing Amina how the difference of squares method can be used to multiply certain numbers without a calculator. He showed her the following: $29 \times 31 = (30 - 1)(30 + 1) = (30^2 - 1^2) = (900 - 1) = 899$.
(a)(i) Use the above process to evaluate 48×52.

(a)(ii) Use the above process to evaluate 77×83.

(b) Explain why this process cannot be used *with whole numbers* to determine the product 58×65.

11. *(Multiple Choice)* One factor of $9x^2 - 4$ is
A. $3x - 4$ **B.** $9x - 2$ **C.** $3x + 2$ **D.** $x - 2$

12. *(Numerical Response)* $4x + 5y$ is a factor of the binomial $a^2x^2 - b^2y^2$. The value of $a^2 + b^2$ is _____.

Record your answer left to right. | | | | |

13. Each student below claims to have factored completely. Identify and explain the error, then write the correct factored form (or state that it does not factor).

(a) $4x^2 - 16 = (2x - 4)(2x + 4)$ (b) $9x^2 + 16 = (3x + 4)^2$ (c) $x^2 - 25 = (x - 5)^2$

Lesson 5 Assignment *Show your work.*

Part A — Tiles, Area Boxes and Grouping (Questions 1–4)

1. Use algebra tiles to factor $5x^2 + 11x + 2$. State the dimensions of the rectangle formed and write the trinomial in factored form.

2. Complete an area box for each trinomial, then write it in factored form.

(a) $2x^2 + 11x + 12$

$2x^2$	
	12

(b) $6x^2 + x - 12$

$6x^2$	
	-12

(c) $4x^2 - 8x + 3$

$4x^2$	
	3

3. Factor the following polynomials by grouping, and verify by expanding.

(a) $4x^2 + 2x + 10x + 5$

(b) $15x^2 + 6x - 10x - 4$

4. *(Explain)* (a) Expand $(2x + 5)(3x - 1)$ *without* collecting the two middle terms. (b) Show that the two middle coefficients multiply to ac for the trinomial you get. (c) Explain why, for *any* factorable $ax^2 + bx + c$, the decomposition pair must have product ac.

Part B — The Method of Decomposition (Questions 5–10)

5. Factor using the method of decomposition.

(a) $8x^2 + 18x + 9$

(b) $6x^2 + 7x + 2$

(c) $10x^2 + 13x - 3$

(d) $8a^2 - 10a - 3$

6. Factor.
(a) $4a^2 + 4a - 3$

(b) $6x^2 - 19x - 7$

(c) $3p^2 - 16p + 5$

(d) $8x^2 - 14x + 3$

7. Factor. *(Watch for a leading negative term, and for a common factor.)*
(a) $9 + 21y + 10y^2$

(b) $14 - 5x - 6x^2$

(c) $6 + 7a - 20a^2$

(d) $20a^2 + 40a - 25$

8. Factor.
(a) $6x^2 + 7xy + 2y^2$

(b) $4m^2 - 4mn - 15n^2$

9. Factor completely. Use a systematic factor-pair list (divide on the calculator) when ac is large. Verify (a) by substituting $x = 2$ into both forms.
(a) $12x^2 - 7x - 12$ (b) $6x^2 - x - 15$ (c) $18x^2 + 3x - 6$ (d) $10x^2 - 11x - 6$

(e) $24x^2 + 2x - 15$ (f) $20x^2 - x - 12$ (g) $-8x^2 + 14x + 15$ (h) $36x^2 + 5x - 24$

10. Identify and explain the error in each factorization. Then write the correct factored form.

(a) $3x^2 - 5x - 2 = (3x - 1)(x + 2)$

(b) $4x^2 + 10x + 6 = (4x + 6)(x + 1)$, fully factored

(c) $8x^2 - 2x - 3 = 8x^2 + 4x - 6x - 3 = 4x(2x + 1) - 3(2x - 1)$, so it does not factor.

Part C — Problem Solving (Questions 11–12)

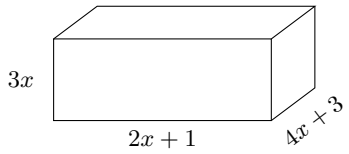
11. A rectangular flower bed has area $3a^2 + 5a - 2$ m^2.

(a) Write the area as the product of two binomials.

(b) The bed is to be completely surrounded by a path 1 m wide. Find and simplify an expression for the area of the path.

(c) The path is poured concrete to a depth of 10 cm. Calculate the volume (in m^3) of concrete used if $a = 6$.

12. A tank in the shape of a rectangular prism has dimensions $3x$, $2x + 1$, and $4x + 3$ metres. A wooden block is dropped into the tank and the tank is then filled with water. The volume of water is $16x^3 + 24x^2 + 8x$ cubic metres.



(a) Determine a polynomial expression for the volume of the wooden block.

(b) Factor the expression in (a) to determine the dimensions of the wooden block.

Part D — Multiple Choice and Numerical Response (Questions 13–16)

13. *(Multiple Choice)* One factor of $18x^2 + 21x - 9$ is

A. $3x + 1$ **B.** $3x - 1$ **C.** $2x - 3$ **D.** $9x + 3$

14. *(Multiple Choice)* The polynomials $6x^2 + 7x - 3$ and $4x^2 - 9$ have in common a factor of

A. $2x + 3$ **B.** $2x - 3$ **C.** $3x - 1$ **D.** $3x + 1$

15. *(Numerical Response)* The factored form of $5x^2 + 18x - 8$ is $(x + a)(bx + c)$ where a, b, and c are integers. The value of b^c, to the nearest hundredth, is _____.

Record your answer left to right. | | | | |

16. *(Numerical Response)* The expression $12x^2 + 13x - 35$ can be written in the form $(ax - b)(cx + d)$ where a, b, c, and d are all positive integers.
Write the value of a in the first box. Write the value of b in the second box.
Write the value of c in the third box. Write the value of d in the fourth box.

Record your answer left to right. | | | | |

Lesson 6 Assignment *Show your work.*

Part A — Perfect Squares, Disguised Trinomials and Differences of Squares (Questions 1–8)

1. Which of the following are perfect square trinomials?

(a) $a^2 + 18a + 81$ (b) $x^2 - 13x + 30$ (c) $9x^2 - 6x + 1$

(d) $49y^2 + 70y + 100$ (e) $a^2 + 16a - 16$ (f) $4x^2 - 4x + 1$

2. Fill in each blank so that the trinomial is a perfect square trinomial.

(a) $x^2 + _____ + 64$ (b) $x^2 - _____ + 121$ (c) $16x^2 + _____ + 25$

(d) $9m^2 + 42m + _____$ (e) $\frac{1}{9}a^2 + _____ + 1$ (f) $_____ - 44y + 4y^2$

3. Factor each perfect square trinomial. Verify (a) by expanding.

(a) $25x^2 - 10x + 1$ (b) $4 + 12x + 9x^2$ (c) $9a^2 - 24ab + 16b^2$ (d) $9x^2 - 66x + 121$

4. Factor completely.

(a) $x^4 + 7x^2 + 12$ (b) $x^4 - 7x^2 + 12$ (c) $a^4 - 13a^2 + 36$

5. Factor completely.

(a) $3x^4 + 4x^2 - 4$ (b) $5a^2b^2 - 19ab + 12$ (c) $9x^2y^2 + 3xy - 20$

6. Factor completely.

(a) $x^4 - 81$ (b) $a^4 - 625b^4$

(c) $3z^4 - 48$

(d) $32x^4 - 2y^4$

7. Return to Lesson 1, Question 9(b). Factor $-12p^3 - 24p^2 - 12p$ completely.

8. Identify and explain the error in each factorization. Then write the correct factored form (or state that it does not factor).
(a) $4x^2 - 12x + 9 = (2x + 3)^2$

(b) $x^4 - 16 = (x^2 - 4)(x^2 + 4)$, fully factored

(c) $x^2 + 6x + 36 = (x + 6)^2$

Part B — Multiple Choice and Numerical Response (Questions 9–14)

9. *(Multiple Choice)* From the expressions below, the one which does **not** represent a perfect square trinomial is

A. $x^2 - 18x + 81$ **B.** $121 + 22x + x^2$ **C.** $9x^2 - 24x + 64$ **D.** $16x^4 + 40x^2 + 25$

10. *(Multiple Choice)* The complete factorization of $n^4 + 25 - 26n^2$ is
A. $(n^2 - 1)(n^2 - 25)$ **B.** $(n^2 + 1)(n^2 + 25)$
C. $n^2(n + 1)(n + 25)$ **D.** $(n + 1)(n - 1)(n + 5)(n - 5)$

11. *(Multiple Choice)* One factor of $z^4 - 256$ is
A. $z + 4$ **B.** $z + 16$ **C.** $z^2 - 4$ **D.** $z^2 + 4$

12. *(Multiple Choice)* One factor of $x^4 - 10x^2 + 9$ is
A. $x + 1$ **B.** $x^2 + 9$ **C.** $x + 9$ **D.** $x - 9$

13. *(Numerical Response)* The polynomial expression $\frac{4}{9}x^2 + \frac{2}{3}x + \frac{1}{4}$ can be written in the form $(Ax + B)^2$. The value of the product AB, to the nearest one hundredth, is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* Triangle PQR is right-angled at P. The area of the triangle is $\frac{5}{2}x^2 + 8x + 6$ cm^2, where x is a positive integer, and the lengths of PR and PQ are binomials in x with integer coefficients. Given that the length of PQ is 16 cm more than the length of PR, the length of QR, to the nearest tenth of a cm, is _____.

Record your answer left to right. | | | | |

Part C — Extension: Higher-Degree Disguises (Questions 15–16)

15. *(Extension)* Factor completely.

(a) $t^6 - 2t^3 - 15$

(b) $z^{10} + 3z^5 - 10$

16. *(Extension)* Factor the polynomial expression completely: $9a^8 - 37a^4 + 4$.

Lesson 7 Assignment *Show your work.*

Part A — Choosing and Using a Strategy (Questions 1–6)

1. Factor completely. Verify (b) by expanding.
(a) $x^2 - 121$ (b) $x^2 - 3x - 40$ (c) $12x^2 + 27$

(d) $b^3 + b^2 + b + 1$ (e) $3q^2 - 5q - 12$ (f) $w^2 + 9w + 18$

2. Factor completely.
(a) $5x^2 - 17x - 12$ (b) $4 + x - 3x^2$ (c) $m^2 - 100n^2$

(d) $75 - 3z^2$ (e) $x^4 + 7x^2 + 10$ (f) $6v^2 - 15v - 36$

3. Factor completely.
(a) $mn + 3m - 4n - 12$ (b) $49a^2 + 84a + 36$ (c) $5 + 14g + 8g^2$

(d) $x^5 - 16x$ (e) $-81 + t^4$

4. Factor completely.
(a) $x^2 - 7xy - 60y^2$ (b) $4a^2 + 17ab + 15b^2$

5. *(Explain your strategy)* For each polynomial, name the method you would use *first* and explain why. Then factor completely.
(a) $2x^3 - 50x$ (b) $6x^2 + 7x - 5$ (c) $ab - 2a + 3b - 6$

6. Identify and explain the error in each factorization. Then write the correct factored form (or state that it does not factor).
(a) $3x^2 - 27 = 3(x^2 - 9)$, fully factored (b) $x^2 + 9 = (x + 3)(x - 3)$ (c) $-2x^2 + 8x + 10 = -2(x^2 + 4x + 5)$

Part B — Multiple Choice and Numerical Response (Questions 7–8)

7. *(Multiple Choice)* $6x^2 + 7x - 20$ has a factor of
A. $2x - 5$ **B.** $3x + 4$ **C.** $2x + 5$ **D.** $x - 4$

8. *(Numerical Response)* If $16a^2 + ka + 49$ is a perfect square trinomial and $k > 0$, then the value of k is _____.
Record your answer left to right. | | | | |

Part C — Extension (Question 9)

9. *(Extension)* Factor completely.
(a) $t^6 - t^3 - 12$ (b) $10t^6 - 7t^3p^2 - 12p^4$

Lesson 8 Assignment *Show your work.*

AN5 — Beginning (BEG)

Questions 1–6: common factors, $x^2 + bx + c$, patterns

1. *(Multiple Choice, BEG)* One factor of $20x^4 - 15x^3 + 10x^2$ is
A. $20x^4$ **B.** $5x^2 - 3x + 2$
C. $4x^2 - 3x + 2$ **D.** $4x^2 - 3x + 1$

2. *(Multiple Choice, BEG)* When fully factored, $s^4t^3 - s^3t^4$ is written
A. $st^2(s - st)$ **B.** $s^2t(st - t)$
C. $s^3t^3(s - t)$ **D.** $s^4t^3(1 - t)$

3. *(Multiple Choice, BEG)* The expression $6uv^2 - 12u^3v + 6uv$, when fully factored, is
A. $uv(6v - 12u^2 + 6)$ **B.** $6u(v^2 - 2u^2v + v)$
C. $6uv(v - 2u^2 + 1)$ **D.** none of these

4. *(Multiple Choice, BEG)* One factor of $p^2 - 6p - 27$ is
A. $p - 3$ **B.** $p - 9$
C. $p + 9$ **D.** $p - 27$

5. *(Numerical Response, BEG)* When the greatest common factor is removed from the binomial $60x - 45x^2$, the binomial can be written in the form $ax(b + cx)$. The value of $a - b - c$ is _____.

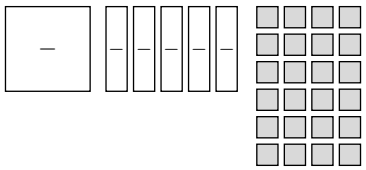
Record your answer left to right. ☐☐☐☐

6. *(BEG)* Factor completely.
(a) $4cd^2 - 8c^3d + 4cd$ (b) $k^2 - 4k - 45$ (c) $25m^2 - 36$

AN5 — Progressing (PRG)

Questions 7–14: $ax^2 + bx + c$, models, factoring completely

7. *(Multiple Choice, PRG)* The algebra tiles below represent an expression (tile key from Lesson 3: white tiles with a minus sign are negative). The factored form of the expression is



A. $(3 - x)(8 + x)$ **B.** $(3 - x)(8 - x)$ **C.** $(8 - x)(3 + x)$ **D.** $24 - 5x - x^2$

8. *(Multiple Choice, PRG)* For which of the following polynomials is $n + 4$ **not** a factor?
A. $n^2 + 4n$ **B.** $n^3 - 16n$
C. $n^2 + n - 12$ **D.** $n^2 - 2n - 35$

9. *(Multiple Choice, PRG)* One factor of $5a^2 - 20a^4$ is
A. $1 - 2a$ **B.** $20a^4$
C. $1 - 4a$ **D.** $1 - a$

10. *(Multiple Choice, PRG)* Which of the following is a factor of $8y^2 - 2y - 1$?
A. $8y - 1$ **B.** $4y - 1$
C. $4y + 1$ **D.** $2y + 1$

11. *(Multiple Choice, PRG)* Which two binomials are **both** factors of $3t^2 - 5t - 2$?
A. $3t - 1$ and $t + 2$ **B.** $3t - 1$ and $t - 2$
C. $t - 2$ and $3t + 1$ **D.** $t + 2$ and $3t + 1$

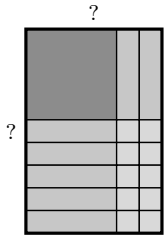
12. *(Numerical Response, PRG)* The expression $3x^2 - 18x - 5x + c$ can be factored by grouping into the product of two binomials. If one of these binomials is $x - 6$, then the value of c is _____.

Record your answer left to right. | | | | |

13. *(PRG)* Factor completely. Verify (b) by expanding.

(a) $8x^4 - 12x^3 + 4x^2$ (b) $10x^2 - x - 3$ (c) $-3x^2 + 12x + 36$

14. *(PRG)* The algebra tiles below have been arranged into a rectangle.



(a) What binomial multiplication does the rectangle represent?

(b) Write the polynomial the tiles represent (expanded form) and its factored form.

AN5 — Mastery (MAS) *Questions 15–21: reasoning, unknown coefficients, errors, several steps*

15. *(Multiple Choice, MAS)* The expression $x^2 + px + 24$ **cannot** be factored over the integers if p has the value

A. -10 **B.** -14

C. 11 **D.** 16

16. *(Numerical Response, MAS)* The largest value of w for which $x^2 - wx + 55$ can be factored over the integers is _____.

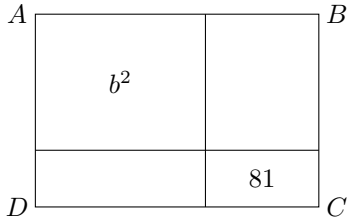
Record your answer left to right. | | | | |

17. *(Numerical Response, MAS)* If $25x^2 - kx + 16$ is a perfect square trinomial and $k > 0$, then the value of k is _____.

Record your answer left to right. | | | | |

18. *(MAS)* Factor completely: $81a^4 - 16$.

19. *(MAS)* Rectangle $ABCD$ has been subdivided into four regions, as shown. The areas of two of the regions are b^2 and 81, and the combined area of the other two regions is $18b$. The region with area b^2 is a square. Determine the perimeter of rectangle $ABCD$, in terms of b.



20. *(MAS)* Identify and explain the error in each factorization. Then write the correct factored form.
(a) $6x^2 - x - 2 = (3x + 2)(2x - 1)$ (b) $x^4 - 81 = (x^2 - 9)^2$

21. *(Written Response, MAS)* Students are investigating polynomials of the form $x^2 + bx + c$, where b and c are integers, with $c = 20$.
(a) State a polynomial with $c = 20$ which **can** be factored over the integers.

(b) State a polynomial with $c = 20$ which **cannot** be factored over the integers. **Explain** why factoring is not possible in this case.

(c) Determine how many polynomials of the form $x^2 + bx + 20$ (with b a nonzero integer) can be factored over the integers.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. $20a^3 = 2\times2\times5\times a\times a\times a$, $32a^2 = 2\times2\times2\times2\times2\times a\times a$; GCF $= 4a^2$. **2.** $12xy^4 = 2\times2\times3\times x\times y\times y\times y\times y$, $18x^2y^3 = 2\times3\times3\times x\times x\times y\times y\times y$; GCF $= 6xy^3$. **3.** (a) $3m$ (b) $2x$ (c) ab^2 (d) mn (e) $3x^2$ (f) 5 (g) $6pq$ (h) $-6x^2z^3$ (i) $2a$ (j) $6n$ (k) $-3ab$ (l) $-7q$. **4.** (a) 5 (b) $3p$ (c) $2x$ (d) 8 (e) $3cd$ (f) $6y$. **5.** (a) 4 (b) $8p$ (c) $2x-3$ (d) $2+3ab$ (e) $4+3y$ (f) $4xmn^2-3n-1$. **6.** (a) $8(m+n)$ (b) $9(xy^2+9)$ (c) $7(3pq-1)$ (d) $5(2c+3d)$ (e) $z(x+1)$ (f) $4x(2x-3)$ (g) $4a(3b-4c)$ (h) $27y^2(2-3y^3)$. **7.** (a) $5(3x-2y+5z)$ (b) $4p(2q+3r-5)$ (c) $t^2(t^2+t+1)$ (d) $7x(x-2y-4z)$ (e) $3ab(2c-d+3e)$ (f) $7a^2b^2(3+4a-6b)$. **8.** (a) $3x^2+4x = x(3x+4)$ (b) $6y^2-9y = 3y(2y-3)$ (c) $-ab^3+2a^3b+6a^2b^2 = ab(2a^2+6ab-b^2)$ (d) $4x^3-6x^2+7x = x(4x^2-6x+7)$. **9.** (a) $15x^2(2x-3)$ (b) $-12p(p^2+2p+1)$ (in Lesson 6: $= -12p(p+1)^2$). **10.** (a) 1020.39 cm^2 (b) $A = 2\pi r(r+h)$ (c) 1020.39 cm^2 (d) (c) is simpler. **11.** (a) $h = 5t(8-t)$ (b) 0, 35, 60, 75, 80, 75, 60, 35, 0 (c) the path is symmetric about the peak at $t=4$: at $t=3$ the rocket is rising, at $t=5$ it has fallen back the same distance (d) $h=-45$; the rocket landed at $t=8$, so it cannot be 45 m below ground. **12.** (a) $2m(2+\pi)$ cm (b) $m^2(4+\pi)$ cm^2. **13.** (a) not the greatest common factor; $4x^2(3x-2)$ (b) sign error; $-3m(2m+3)$ (c) the 1 is missing; $2x(2y+1)$. **14.** C. **15.** D. **16.** B. **17.** 14. **18.** 21. **19.** (a) $\frac{1}{3}x(1-3x)$ (b) $\frac{1}{4}a(20a+3)$ (c) $-\frac{1}{5}y(y+5)$. **20.** (a) $\frac{1}{7}$ (b) $\frac{1}{6}x$ (c) $0.2a$. **21.** (a) $\frac{1}{3}(1+x)$ (b) $\frac{1}{4}(m-5)$ (c) $\frac{1}{12}h(8a-9b)$ (d) $\frac{3}{4}(x+12)$ (e) $\frac{1}{2}(5x^2-2)$ (f) $\frac{1}{3}(3z^2-z+2)$ (g) $\frac{5}{3}\pi r^2(r-3)$ (h) $\frac{1}{4}h(A+4\pi r^2)$. **22.** (a) $s = \frac{1}{2}t(2v+at)$ (b) 4.25 m. **23.** (a) the common factor is $\frac{1}{8}x$, not $\frac{1}{4}x$; $\frac{1}{8}x(2x-1)$ (b) the second factor should not contain π; $\frac{1}{4}\pi r(r-8)$.

Lesson 2 Assignment

1. (a) $(x+6)(4x-5)$ (b) $(x+3)(9y+4)$ (c) $(x+2)(x-1)$ (d) $(m+n)(x-y)$ (e) $6(a+b)(x-2y)$ (f) $3(a+7)(a-2)$ (g) $5(b+c)(3a+d)$ (h) $(4x-5)(x+3)$ (i) $(2-9p)(5-4p)$. **2.** (a) $(x+3)(x+7)$ (b) $(x+4)(x+12)$ (c) $(m-6)(m+3)$ (d) $(a-8)(a-6)$ (e) $(x-12)(x-5)$ (f) $(t+9)(t-4)$. **3.** (a) $(x+1)(3x+4)$ (b) $(4x+1)(x+2)$ (c) $(m+4)(2m+3)$ (d) $(2b-3)(5b-4)$ (e) $(a-3)(3a-2)$ (f) $(7x+3)(x-5)$ (g) $(5+p)^2$ (h) $(7-y)(3-y)$ (i) $(b+x)(b+y)$. **4.** (a) $(x-p)(x-q)$ (b) $(b-3)(6b-a)$ (c) $(5x-7)(x-3y^2)$. **5.** (a) sign error: $-8(5-m) = +8(m-5)$; $(m-5)(m+8)$ (b) sign error: $-5x-15 = -5(x+3)$; $(x+3)(2x-5)$ (c) the 1 is missing; $(a-4)(3a+1)$. **6.** C. **7.** B. **8.** A. **9.** 10.

Lesson 3 Assignment

1. (a) x^2+8x+7 (b) length $x+7$, width $x+1$ (c) $(x+1)(x+7)$. **2.** (a) $3, 4$ (b) $1, 8$ (c) $5, 8$ (d) $-5, -8$ (e) $1, 9$ (f) $-2, -7$ (g) $-3, -16$ (h) $-8, -12$. **3.** a) $8, -3$ b) $-8, 3$ c) $4, -6$ d) $9, -5$ e) $-4, -9$ f) $5, -5$. **4.** (a) x^2-3x-4 (b) $x+1$ by $x-4$; $(x+1)(x-4)$ (c) each sits in a negative row and a positive column: $(+1)(-1) = -1$. **5.** (a) 5 (b) $1, 10$ (c) 9 (d) $1, 15$ (e) $(z+10)$ (f) $(b-12)$. **6.** (a) $(x+1)(x+4)$ (b) $(x-1)(x-4)$ (c) $(x-3)(x-9)$ (d) $(x-3)(x-10)$. **7.** (a) $(x+1)(x+12)$ (b) not possible (c) $(y-3)(y-10)$ (d) $(y+3)(y+16)$ (e) $(f-3)(f-8)$ (f) $(p-2)(p-15)$ (g) not possible. **8.** (a) $(x-5)(x+2)$ (b) $(x-8)(x+5)$ (c) $(x+8)(x-5)$ (d) $(x-9)(x+2)$. **9.** (a) $(x-11)(x-2)$ (b) $(x-7)(x+3)$ (c) $(x+9)(x-3)$ (d) not possible (e) $(x-7)(x+2)$ (f) $(x-6)^2$ (g) $(m+21)(m+2)$ (h) $(a-16)(a-3)$. **10.** (a) $2(x+1)(x+4)$ (b) $3(x-6)(x-8)$ (c) $-2(a+6)(a+7)$ (d) $4(x-1)(x-3)$ (e) $a(x-5)(x-6)$ (f) $-6a^2(a-5)(a-6)$. **11.** (a) $2(x+5)(x+3)$ (b) $4(x-7)(x+1)$ (c) $5(x-5)(x-1)$ (d) $-2(a-8)(a+6)$ (e) $b^2(x-10)(x+4)$ (f) $3x(x+5)(x-4)$. **12.** (a) $(x+7)(x+10)$ (b) length 15 m, width 12 m. **13.** $B=1$, $I=2$, $L=3$, $O=4$, $A=5$, $G=6$, $N=7$, $S=8$, $T=9$; the code spells **BILBO BAGGINS**. **14.** (a) signs must be opposite; $(x-7)(x+2)$ (b) right product, wrong sum; $(x+3)(x+4)$ (c) the GCF 2 was dropped; $2(x+2)(x+3)$. **15.** (a) $(x+7y)(x+9y)$ (b) $(x+10y)(x-2y)$ (c) $(a-7b)^2$ (d) $(p-12q)(p-q)$ (e) $(x+10y)(x-9y)$ (f) $(x-25y)(x+3y)$. **16.** (a) $3(x-8y)(x-9y)$ (b) $2(b-16v)(b+10v)$ (c) $4(c+10d)(c-8d)$. **17.** C. **18.** D. **19.** A. **20.** A. **21.** D. **22.** A. **23.** 37.

Lesson 4 Assignment

1. (a) $(y+8)$ (b) $(d-13)$ (c) $(r+s)$. **2.** (a) $(x-9)(x+9)$ (b) $(x-1)(x+1)$ (c) $(x-13)(x+13)$ (d) $(x-30)(x+30)$. **3.** It is the $b=0$ case of x^2+bx+c, with $c=-k^2$; the pair needed is always k and $-k$. **4.** (a) $(p-q)(p+q)$ (b) $(e-9)(e+9)$ (c) $(1-j)(1+j)$ (d) $(f-7g)(f+7g)$ (e) $(4x-9)(4x+9)$ (f) $(5a-2b)(5a+2b)$ (g) sum of squares — not factorable (h) $(8c-11d)(8c+11d)$ (i) $(x-1)(x+1)(x^2+1)$. **5.** (a) $L=(7p+3q)$ ft, $W=(7p-3q)$ ft (b) $28p=252$, $p=9$ (c) 84 ft by 42 ft. **6.** (a) $5(x-3)(x+3)$ (b) $6(a-2y)(a+2y)$ (c) $2(t^2+9s^2)$, not factorable further (d) $3(x-y)(x+y)$ (e) $4(ab-4)(ab+4)$ (f) $3(2-pq)(2+pq)$. **7.** (a) $(xy-4)(xy+4)$ (b) $(a-bc)(a+bc)$ (c) $(9x-yz)(9x+yz)$ (d) $(mn-pq)(mn+pq)$ (e) $(4xy-1)(4xy+1)$ (f) $(ab-3cd)(ab+3cd)$. **8.** (a) $A=\pi R^2-\pi r^2$ (b) $\pi(R-r)(R+r)$ (c) 50π. **9.** (a) $\frac{1}{2}m(v-u)(v+u)$ (b) 6400. **10.** (a)(i) 2496 (ii) 6391 (b) their average 61.5 is not a whole number, so 58×65 is not a difference of two whole-number squares. **11.** C. **12.** 41 ($a=4$, $b=5$, $a^2+b^2=16+25=41$). **13.** (a) not complete; $4(x-2)(x+2)$ (b) a sum of squares does not factor; $(3x+4)^2 = 9x^2+24x+16$ (c) $(x-5)^2$ has a middle term; $(x-5)(x+5)$.

Lesson 5 Assignment

1. $(5x+1)(x+2)$; rectangle dimensions $x+2$ by $5x+1$. **2.** (a) $(2x+3)(x+4)$ (b) $(3x-4)(2x+3)$ (c) $(2x-3)(2x-1)$. **3.** (a) $(2x+1)(2x+5)$ (b) $(5x+2)(3x-2)$. **4.** (a) $6x^2-2x+15x-5$ (b) $(-2)(15) = -30 = ac$ (c) $(px+q)(rx+s)$ has middle pieces ps and qr with $(ps)(qr) = (pr)(qs) = ac$. **5.** (a) $(4x+3)(2x+3)$ (b) $(2x+1)(3x+2)$ (c) $(2x+3)(5x-1)$ (d) $(2a-3)(4a+1)$. **6.** (a) $(2a+3)(2a-1)$ (b) $(2x-7)(3x+1)$ (c) $(p-5)(3p-1)$ (d) $(2x-3)(4x-1)$. **7.** (a) $(5y+3)(2y+3)$ (b) $-(x+2)(6x-7)$ or $(x+2)(7-6x)$ (c) $-(4a-3)(5a+2)$ or $(3-4a)(5a+2)$ (d) $5(2a+5)(2a-1)$. **8.** (a) $(2x+y)(3x+2y)$ (b) $(2m+3n)(2m-5n)$. **9.** (a) $(3x-4)(4x+3)$ (b) $(3x-5)(2x+3)$ (c) $3(3x+2)(2x-1)$ (d) $(2x-3)(5x+2)$ (e) $(6x+5)(4x-3)$ (f) $(4x+3)(5x-4)$ (g) $-(2x-5)(4x+3)$ (h) $(9x+8)(4x-3)$. **10.** (a) wrong signs; $(3x+1)(x-2)$ (b) GCF left inside; $2(2x+3)(x+1)$ (c) sign error, $-6x-3 = -3(2x+1)$; $(2x+1)(4x-3)$. **11.** (a) $(a+2)(3a-1)$ (b) $8a+6$ m^2 (c) 5.4 m^3. **12.** (a) $8x^3+6x^2+x$ (b) x, $2x+1$, $4x+1$ (metres). **13.** B. **14.** A. **15.** 0.04 (recorded .04). **16.** $a=4$, $b=5$, $c=3$, $d=7$ (i.e. 4537).

Lesson 6 Assignment

1. (a), (c), (f) are perfect square trinomials. **2.** (a) $16x$ (b) $22x$ (c) $40x$ (d) 49 (e) $\frac{2}{3}a$ (f) 121. **3.** (a) $(5x-1)^2$ (b) $(3x+2)^2$ (c) $(3a-4b)^2$ (d) $(3x-11)^2$. **4.** (a) $(x^2+3)(x^2+4)$ (b) $(x^2-3)(x-2)(x+2)$ (c) $(a-3)(a+3)(a-2)(a+2)$. **5.** (a) $(x^2+2)(3x^2-2)$ (b) $(ab-3)(5ab-4)$ (c) $(3xy+5)(3xy-4)$. **6.** (a) $(x-3)(x+3)(x^2+9)$ (b) $(a-5b)(a+5b)(a^2+25b^2)$ (c) $3(z-2)(z+2)(z^2+4)$ (d) $2(2x-y)(2x+y)(4x^2+y^2)$. **7.** $-12p(p+1)^2$. **8.** (a) sign of the middle term; $(2x-3)^2$ (b) not complete; $(x-2)(x+2)(x^2+4)$ (c) not a perfect square (middle would be $12x$); does not factor. **9.** C. **10.** D. **11.** A. **12.** A. **13.** 0.33 (recorded .33). **14.** 21.6 cm. **15.** (a) $(t^3-5)(t^3+3)$ (b) $(z^5+5)(z^5-2)$. **16.** $(3a^2-1)(3a^2+1)(a^2-2)(a^2+2)$.

Lesson 7 Assignment

1. (a) $(x-11)(x+11)$ (b) $(x-8)(x+5)$ (c) $3(4x^2+9)$ (d) $(b+1)(b^2+1)$ (e) $(q-3)(3q+4)$ (f) $(w+6)(w+3)$. **2.** (a) $(x-4)(5x+3)$ (b) $-(3x-4)(x+1)$ or $(4-3x)(x+1)$ (c) $(m-10n)(m+10n)$ (d) $3(5-z)(5+z)$ (e) $(x^2+5)(x^2+2)$ (f) $3(2v+3)(v-4)$. **3.** (a) $(n+3)(m-4)$ (b) $(7a+6)^2$ (c) $(4g+5)(2g+1)$ (d) $x(x-2)(x+2)(x^2+4)$ (e) $(t-3)(t+3)(t^2+9)$. **4.** (a) $(x-12y)(x+5y)$ (b) $(a+3b)(4a+5b)$. **5.** (a) common factor, then difference of squares; $2x(x-5)(x+5)$ (b) decomposition ($a \neq 1$); $(3x+5)(2x-1)$ (c) grouping (four terms); $(b-2)(a+3)$. **6.** (a) not complete; $3(x-3)(x+3)$ (b) a sum of squares does not factor (c) signs not flipped; $-2(x-5)(x+1)$. **7.** C ($6x^2+7x-20=(2x+5)(3x-4)$). **8.** 56 ($16a^2+56a+49=(4a+7)^2$). **9.** (a) $(t^3-4)(t^3+3)$ (b) $(2t^3-3p^2)(5t^3+4p^2)$.

Lesson 8 Assignment

1. C. **2.** C. **3.** C. **4.** B. **5.** 14 ($15x(4-3x)$, so $15-4+3=14$). **6.** (a) $4cd(d-2c^2+1)$ (b) $(k-9)(k+5)$ (c) $(5m-6)(5m+6)$. **7.** A. **8.** D. **9.** A. **10.** C. **11.** C. **12.** 30 ($3x^2-23x+30=(x-6)(3x-5)$). **13.** (a) $4x^2(2x-1)(x-1)$ (b) $(5x-3)(2x+1)$ (c) $-3(x-6)(x+2)$. **14.** (a) $(x+2)(x+5)$ (b) $x^2+7x+10=(x+2)(x+5)$. **15.** D. **16.** 56 ($x^2-56x+55=(x-1)(x-55)$). **17.** 40 ($25x^2-40x+16=(5x-4)^2$). **18.** $(3a-2)(3a+2)(9a^2+4)$. **19.** $4b+36$. **20.** (a) signs in the wrong brackets; $(3x-2)(2x+1)$ (b) $(x^2-9)^2$ has a middle term and x^2-9 factors again; $(x-3)(x+3)(x^2+9)$. **21.** (a) e.g. $x^2+9x+20=(x+4)(x+5)$ (b) e.g. $x^2+3x+20$ — no two integers multiply to 20 and add to 3 (c) 6 polynomials, $b=\pm 9, \pm 12, \pm 21$.